

Table S1. Virtual screening of the sixty designed compounds by the three-classifier GROVER ensemble, with probabilities reported as percentages.

Each of the sixty designed compounds (ten series, **I–X**, of six members each) was encoded with the fine-tuned GROVER representation and scored independently by three classifiers: a random forest (RF), XGBoost and a multilayer perceptron (MLP). For each classifier, *Active Probability (%)* is the probability assigned to the Active class, expressed as a percentage, and *Predicted Label* applies a decision threshold of 50%. *Active Hits* is the number of classifiers predicting Active (0–3); a compound was taken as predicted active when a majority, that is at least two of the three, agreed. Series **IX** is the only series in which all six members met this criterion, and the only one in which every member was predicted active by all three classifiers; it was the series carried forward to synthesis.

Code	SMILES	RF		XGBoost		MLP		Active
		Prob. (%)	Label	Prob. (%)	Label	Prob. (%)	Label	Hits
Series I								
Ia	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=C(C=CC=C4)C1)=O)=CC=C32)C=C1)NO</chem>	54.50	Active	90.82	Active	2.69	Inactive	2
Ib	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=CC(C1)=CC=C4)=O)=CC=C32)C=C1)NO</chem>	52.50	Active	78.02	Active	1.17	Inactive	2
Ic	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=CC=C(C=C4)F)=O)=CC=C32)C=C1)NO</chem>	42.00	Inactive	41.17	Inactive	0.03	Inactive	0
Id	<chem>O=C(CCC1=CC=C(CN2C3=CC(NC(C4=C(C=CC=C4)C1)=O)=CC=C3N=C2)C=C1)NO</chem>	54.00	Active	85.15	Active	2.23	Inactive	2
Ie	<chem>O=C(CCC1=CC=C(CN2C3=CC(NC(C4=CC(C1)=CC=C4)=O)=CC=C3N=C2)C=C1)NO</chem>	53.00	Active	78.02	Active	1.22	Inactive	2
If	<chem>O=C(CCC1=CC=C(CN2C3=CC(NC(C4=CC=C(C=C4)F)=O)=CC=C3N=C2)C=C1)NO</chem>	42.00	Inactive	34.10	Inactive	0.03	Inactive	0
Series II								
IIa	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=C(C=CC=C4)C1)=O)=CC=C32)C=C1)NO</chem>	37.00	Inactive	42.56	Inactive	0.02	Inactive	0
IIb	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=CC(C1)=CC=C4)=O)=CC=C32)C=C1)NO</chem>	43.50	Inactive	91.50	Active	0.76	Inactive	1
IIc	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=CC=C(C=C4)F)=O)=CC=C32)C=C1)NO</chem>	20.50	Inactive	0.48	Inactive	0.00	Inactive	0
IId	<chem>O=C(CCC1=CC=C(CN2C3=CC(CC(C4=C(C=CC=C4)C1)=O)=CC=C3N=C2)C=C1)NO</chem>	36.50	Inactive	42.56	Inactive	0.02	Inactive	0
IIe	<chem>O=C(CCC1=CC=C(CN2C3=CC(CC(C4=CC(C1)=CC=C4)=O)=CC=C3N=C2)C=C1)NO</chem>	42.50	Inactive	89.51	Active	0.77	Inactive	1
IIIf	<chem>O=C(CCC1=CC=C(CN2C3=CC(CC(C4=CC=C(C=C4)F)=O)=CC=C3N=C2)C=C1)NO</chem>	21.00	Inactive	0.75	Inactive	0.00	Inactive	0
Series III								
IIIa	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=C(C=CC=C4)C1)=CC=C32)C=C1)NO</chem>	40.50	Inactive	8.21	Inactive	0.01	Inactive	0
IIIb	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=CC(C1)=CC=C4)=CC=C32)C=C1)NO</chem>	50.00	Active	64.50	Active	0.05	Inactive	2
IIIc	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=CC=C(C=C4)F)=CC=C32)C=C1)NO</chem>	35.00	Inactive	1.67	Inactive	0.00	Inactive	0
IIId	<chem>O=C(CCC1=CC=C(CN2C3=CC(CCC4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)NO</chem>	39.50	Inactive	17.17	Inactive	0.01	Inactive	0
IIIe	<chem>O=C(CCC1=CC=C(CN2C3=CC(CCC4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)NO</chem>	48.00	Inactive	64.50	Active	0.06	Inactive	1
IIIIf	<chem>O=C(CCC1=CC=C(CN2C3=CC(CCC4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO</chem>	35.50	Inactive	1.00	Inactive	0.00	Inactive	0
Series IV								
IVa	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)NO</chem>	66.00	Active	86.07	Active	33.10	Inactive	2
IVb	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)NO</chem>	76.00	Active	99.55	Active	95.83	Active	3

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Table S1 (continued)

Code	SMILES	RF		XGBoost		MLP		Active
		Prob. (%)	Label	Prob. (%)	Label	Prob. (%)	Label	Hits
IVc	<chem>O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO</chem>	43.50	Inactive	4.02	Inactive	0.01	Inactive	0
IVd	<chem>O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)NO</chem>	66.50	Active	98.21	Active	29.94	Inactive	2
IVe	<chem>O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)NO</chem>	74.50	Active	99.52	Active	95.00	Active	3
IVf	<chem>O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO</chem>	40.50	Inactive	4.02	Inactive	0.01	Inactive	0
<i>Series V</i>								
Va	<chem>O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)=O)NO</chem>	59.00	Active	94.01	Active	57.67	Active	3
Vb	<chem>O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)=O)NO</chem>	69.50	Active	99.37	Active	98.21	Active	3
Vc	<chem>O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO</chem>	28.50	Inactive	2.22	Inactive	0.04	Inactive	0
Vd	<chem>O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)=O)NO</chem>	57.50	Active	95.98	Active	59.92	Active	3
Ve	<chem>O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)=O)NO</chem>	70.50	Active	99.37	Active	98.28	Active	3
Vf	<chem>O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO</chem>	30.50	Inactive	4.73	Inactive	0.04	Inactive	0
<i>Series VI</i>								
VIa	<chem>O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)=O)NO</chem>	83.00	Active	98.95	Active	19.00	Inactive	2
VIb	<chem>O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)=O)NO</chem>	86.00	Active	99.93	Active	90.57	Active	3
VIc	<chem>O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO</chem>	67.00	Active	25.32	Inactive	0.01	Inactive	1
VIId	<chem>O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)=O)NO</chem>	81.00	Active	99.25	Active	16.39	Inactive	2
VIe	<chem>O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)=O)NO</chem>	86.00	Active	99.87	Active	88.46	Active	3
VIIf	<chem>O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO</chem>	66.00	Active	59.48	Active	0.01	Inactive	2
<i>Series VII</i>								
VIIa	<chem>O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)=O)NO</chem>	65.50	Active	92.64	Active	0.13	Inactive	2
VIIb	<chem>O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)=O)NO</chem>	70.50	Active	98.39	Active	4.24	Inactive	2
VIIc	<chem>O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO</chem>	37.00	Inactive	2.02	Inactive	0.00	Inactive	0
VIId	<chem>O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)=O)NO</chem>	62.00	Active	95.32	Active	0.11	Inactive	2
VIIe	<chem>O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)=O)NO</chem>	69.50	Active	97.52	Active	3.13	Inactive	2
VIIIf	<chem>O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO</chem>	34.50	Inactive	2.46	Inactive	0.00	Inactive	0
<i>Series VIII</i>								
VIIIa	<chem>O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)NO</chem>	70.50	Active	83.17	Active	4.26	Inactive	2
VIIIb	<chem>O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)NO</chem>	82.00	Active	99.33	Active	66.45	Active	3
VIIIc	<chem>O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO</chem>	40.50	Inactive	6.57	Inactive	0.00	Inactive	0
VIIIId	<chem>O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)NO</chem>	69.00	Active	83.17	Active	3.57	Inactive	2

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Table S1 (continued)

Code	SMILES	RF		XGBoost		MLP		Active
		Prob. (%)	Label	Prob. (%)	Label	Prob. (%)	Label	Hits
VIIIe	<chem>O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)NO</chem>	81.00	Active	99.66	Active	59.35	Active	3
VIIIIf	<chem>O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=CC(C(C=C4)F)=CC=C3N=C2)C=C1)NO</chem>	39.00	Inactive	7.89	Inactive	0.00	Inactive	0
<i>Series IX</i>								
IXa	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=C(C1)C=CC=C4)=O)=CC=C32</chem>	85.00	Active	99.35	Active	99.94	Active	3
IXb	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=CC(C1)=CC=C4)=O)=CC=C32</chem>	84.50	Active	99.45	Active	99.87	Active	3
IXc	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=CC=C(F)C=C4)=O)=CC=C32</chem>	75.00	Active	65.24	Active	94.37	Active	3
IXd	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=C(C1)C=CC=C4)=O)=CC=C3N=C2</chem>	85.00	Active	99.17	Active	99.93	Active	3
IXe	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=CC(C1)=CC=C4)=O)=CC=C3N=C2</chem>	85.50	Active	99.23	Active	99.88	Active	3
IXf	<chem>O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=CC=C(F)C=C4)=O)=CC=C3N=C2</chem>	76.00	Active	66.49	Active	94.38	Active	3
<i>Series X</i>								
Xa	<chem>O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C32)C=C1)NO</chem>	81.00	Active	99.78	Active	47.43	Inactive	2
Xb	<chem>O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO</chem>	59.50	Active	18.89	Inactive	0.02	Inactive	1
Xc	<chem>O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)C1)=CC=C3N=C2)C=C1)NO</chem>	80.50	Active	99.67	Active	43.63	Inactive	2
Xd	<chem>O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C3N=C2)C=C1)NO</chem>	88.50	Active	99.91	Active	97.08	Active	3
Xe	<chem>O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO</chem>	59.50	Active	14.81	Inactive	0.02	Inactive	1
Xf	<chem>O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(C1)=CC=C4)=CC=C32)C=C1)NO</chem>	88.00	Active	99.93	Active	97.58	Active	3